Tutorium – Business Process Technologies and Management

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General hints

- In most cases the tasks should state clearly what to do
- If you must make assumptions write them down
- If not stated otherwise and not clear from the context a PetriNet is referred to as an E/C Net
- Always label transitions and places

BPMN

Model the following problem statement as a well structured BPMN Model:

A customer comes in a store. The customer picks a drink and something to eat at the same time. After that the customer has to pay for the things he or she wants to buy or can pick something to eat and drink again. If the customer decides to pay the customer can choose to pay by cash or by card. After the payment was successful the customer leaves the store.

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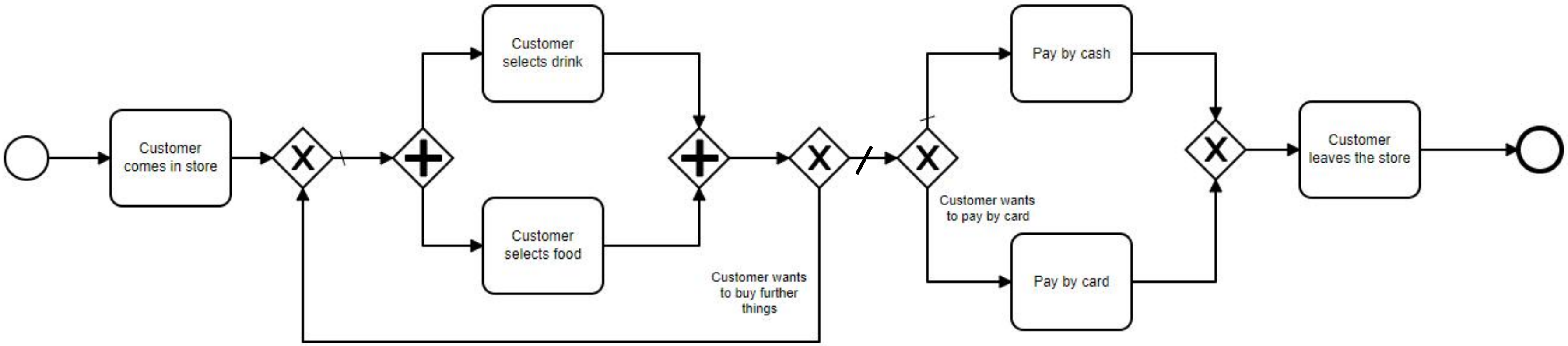
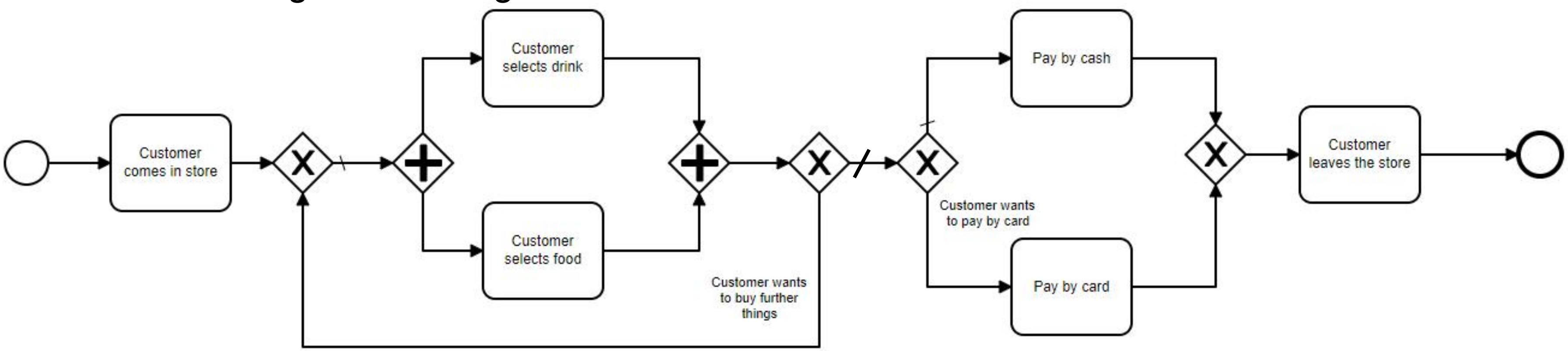
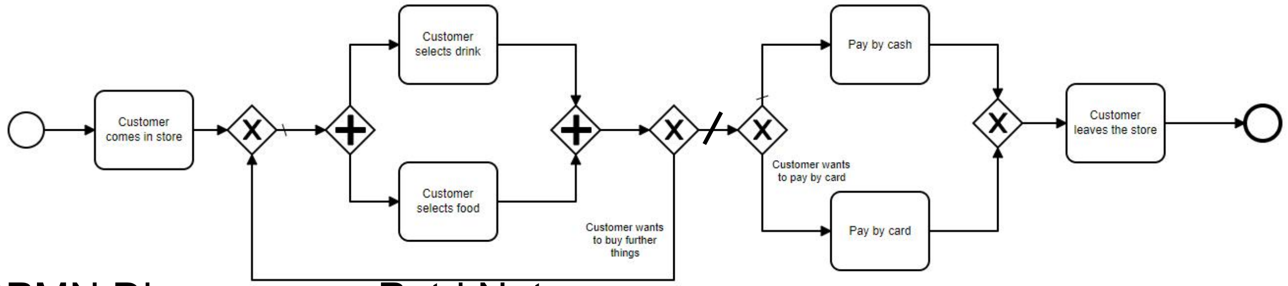


Petri Nets

Draw the following BPMN Diagram as an Petri Net



Petri Nets



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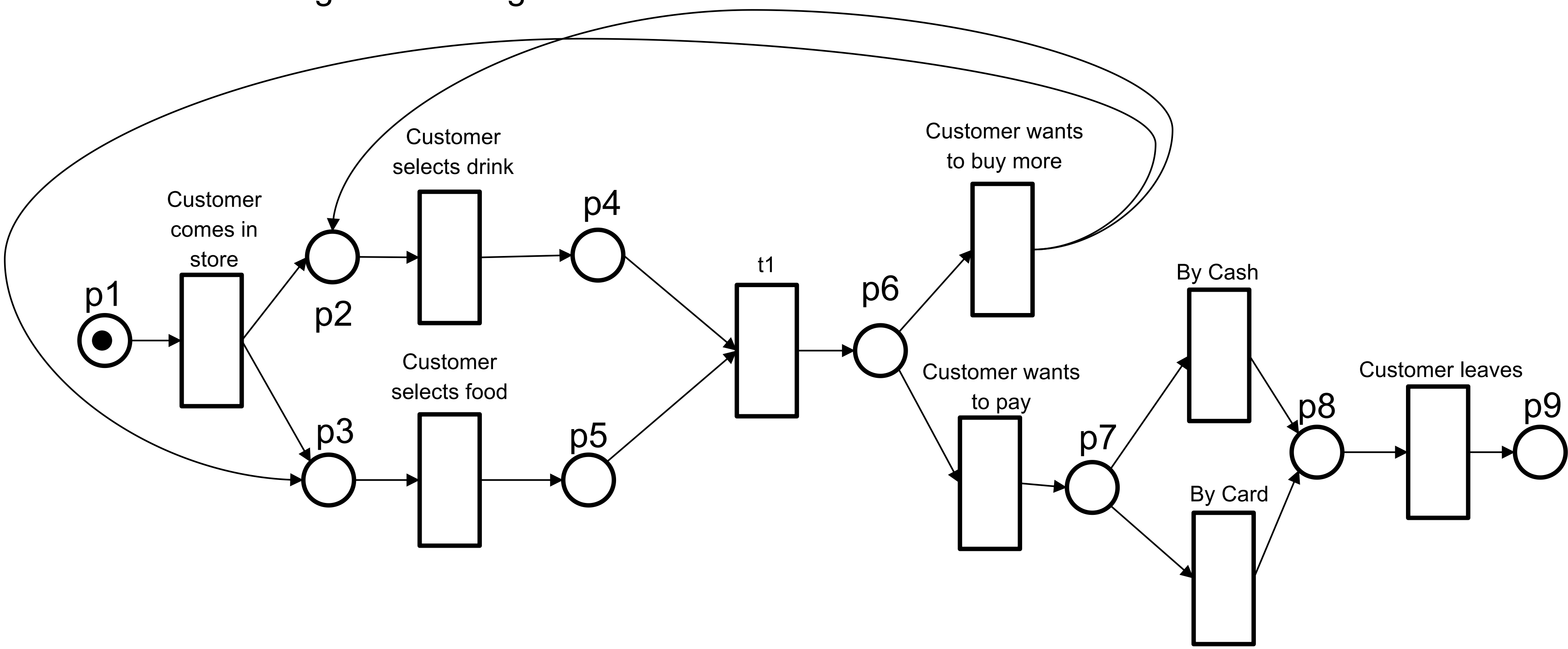


Petri Nets

Explain the difference between an EC Net and and an PT Net.

Petri Nets - Solution

Explain the difference between an EC Net and and an PT Net.

Solution:

Event/Condition Nets (EC Nets): max. 1 token per place

Place/Transition nets: E/C Nets are now extended by edge weights and place capacities,

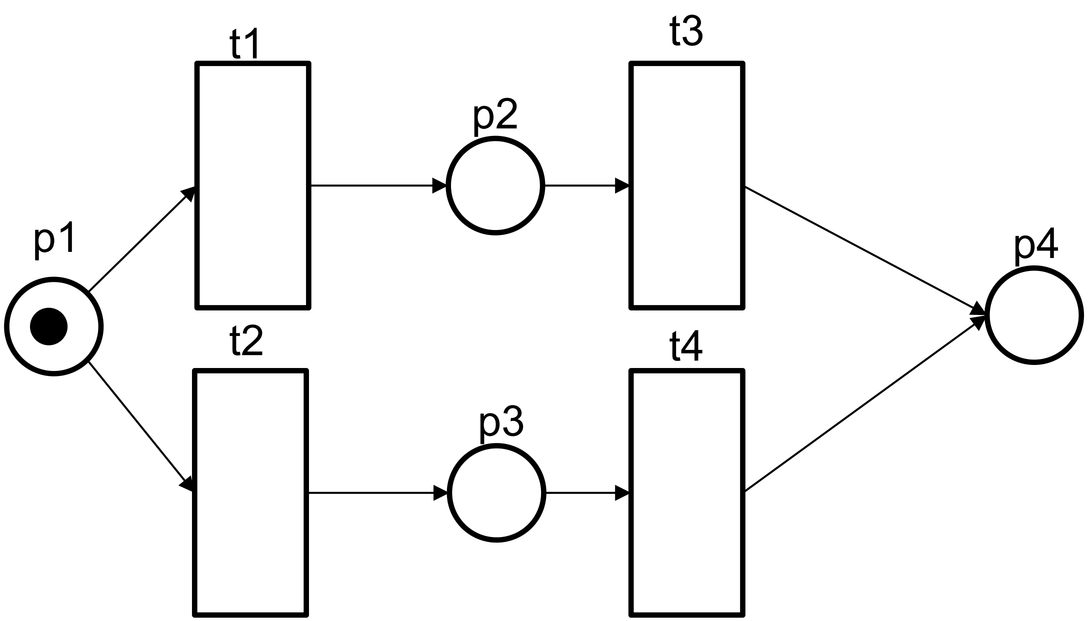
Places can carry more than one token.

Petri Nets

Draw a decision flow from p1 to p4. P1 should initially contain one token, and p4 should finally contain one token. A Maximum of 4 places are allowed.

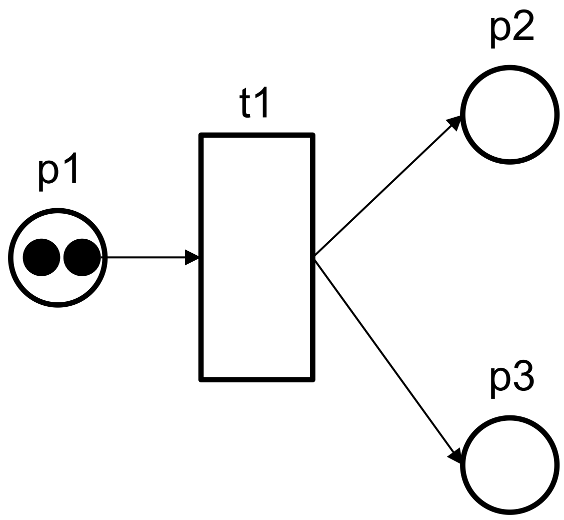
Petri Nets

Draw a parallel flow from p1 to p4. P1 should initially contain one token, and p4 should finally contain one token. A Maximum of 4 places are allowed.



Petri Nets

Ensure that the following net is always live and that there are always exactly two tokens in the net. You are not allowed to add any new places.



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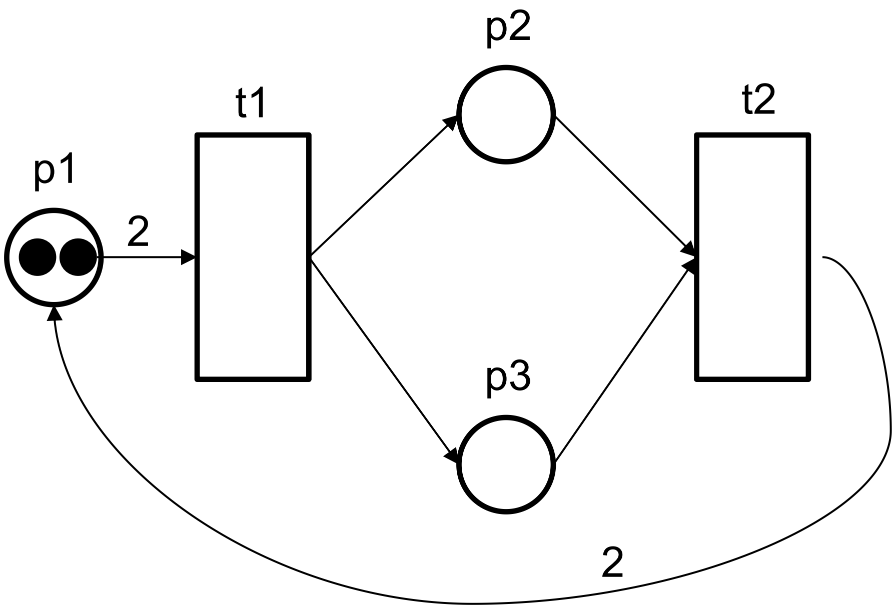
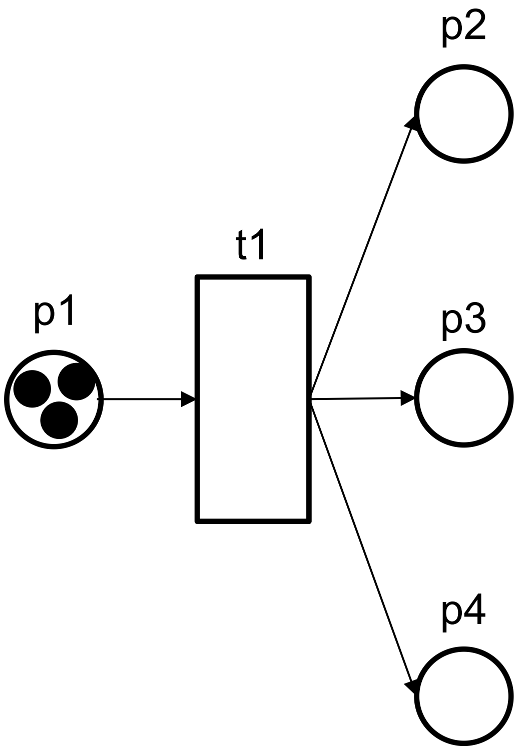


Petri Nets

Make the net exactly 3-bounded and live. You are not allowed to add any new places.



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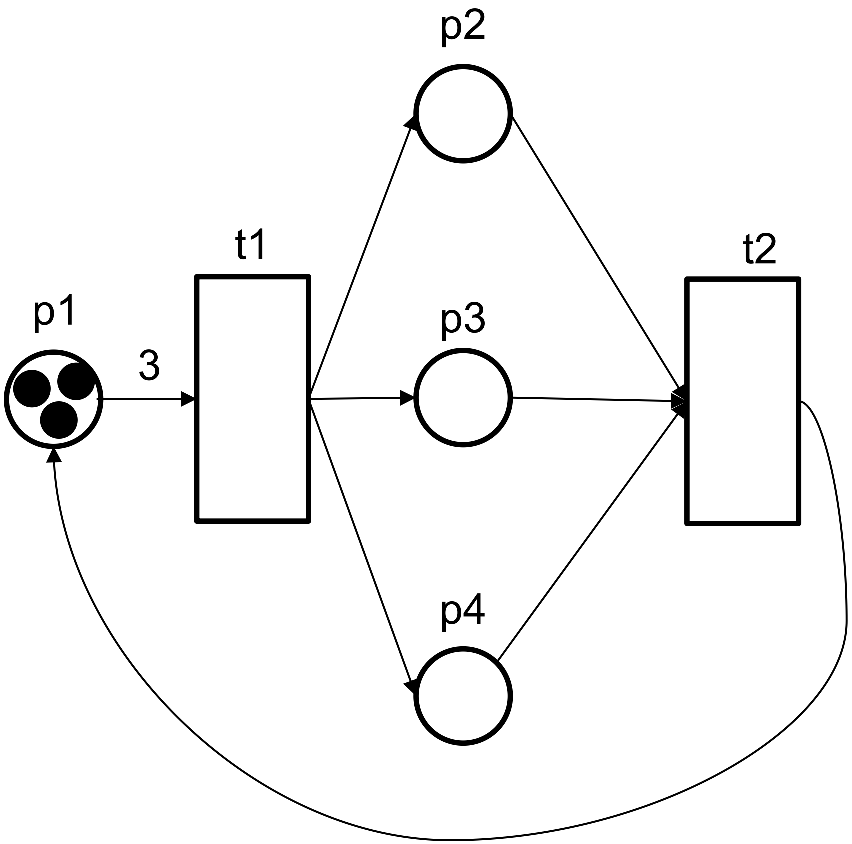
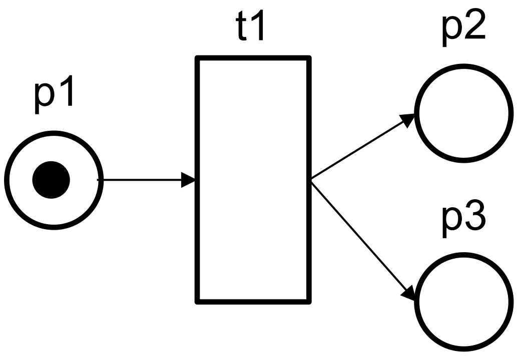


Petri Nets

Introduce a deadlock into the net. You are only allowed to add one place and one transition



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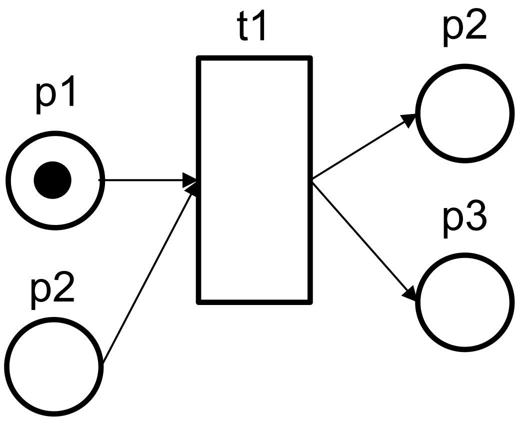


Petri Nets

What type of Petri Net is this and why? Draw the Petri Net next.

$N = \{P,T,F,M_0\}$

$P = \{p1,p2,p3,p4,p5\}$

$T = \{t1,t2,t3,t4,t5\}$

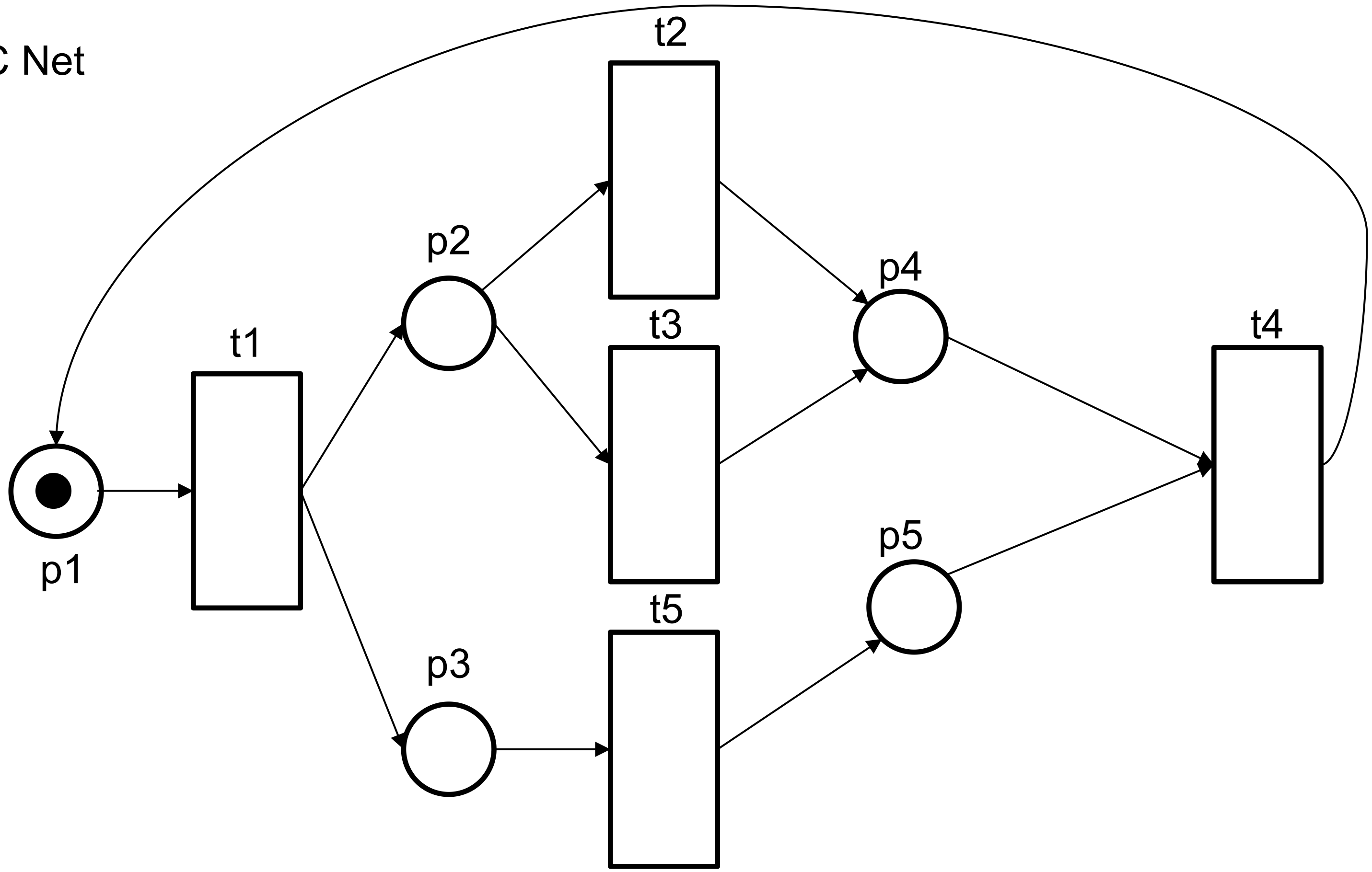
$F = \{\{p1,t1\}, \{p2,t2\}, \{p2,t3\}, \{t1,p2\}, \{t1,p3\}, \{p3,t5\}, \{t2,p4\}, \{t3,p4\}, \{t5,p5\}, \{p4,t4\}, \{p5,t4\}, \{t4,p1\}\}$

$M_0 = \{\{p1,1\}\}$

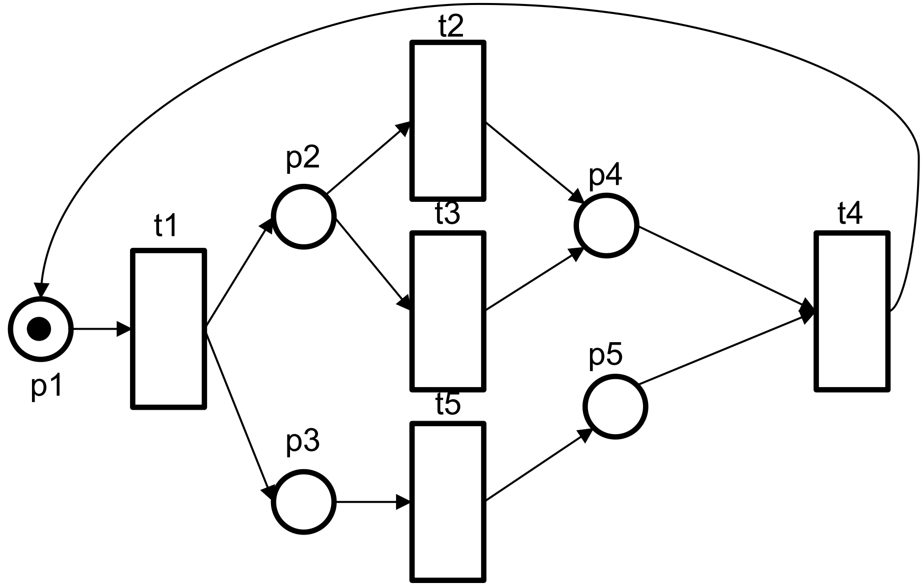
Petri Nets

What type of Petri Net is this and why? Draw the Petri Net next.

EC Net



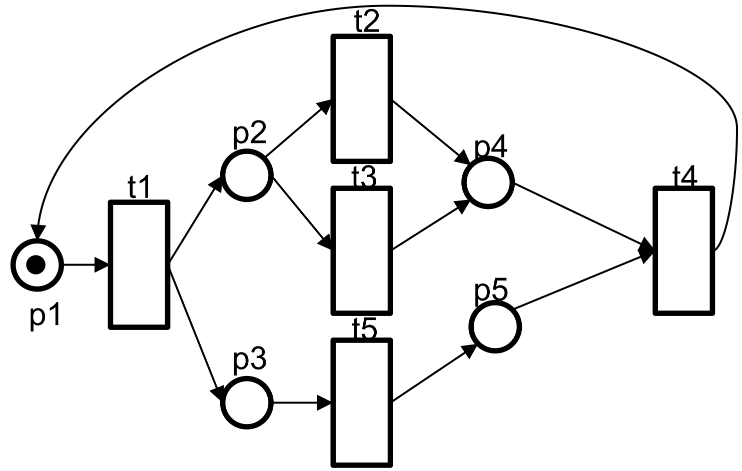
Petri Nets



Perform a reachability analysis for the following net (we have just created this): What is the k of k-bounded?

No.	P1	P2	P3	P4	P5	Firing transitions
0						
1						
2						
3						
4						
5						
6						

Petri Nets



Perform a reachability analysis for the following net (we have just created this). What is the k of k-bounded?

No.	P1	P2	P3	P4	P5	Firing transitions
M0	X					T1->M1
M1		X	X			T2->M2, T3->M2, T5->M3
M2			X	X		T5->M4
M3		X			X	T2->M4,T3->M4
M4				X	X	T4->M0

Petri Nets

What type of Petri Net is this and why? Draw the Petri Net.

N = {P,T,F,C,W,M_0}

P = {p1,p2,p3,p4,p5,p6}

T = {t1,t2,t3,t4}

F = {{p1,t1}, {t1,p2}, {p2,t2}, {t1,p3}, {t1,p4}, {t1,p5}, {p3,t2}, {t2,p6}, {p4,t3}, {t3,p5}, {p5,t4}, {t4,p6}}

C = {}

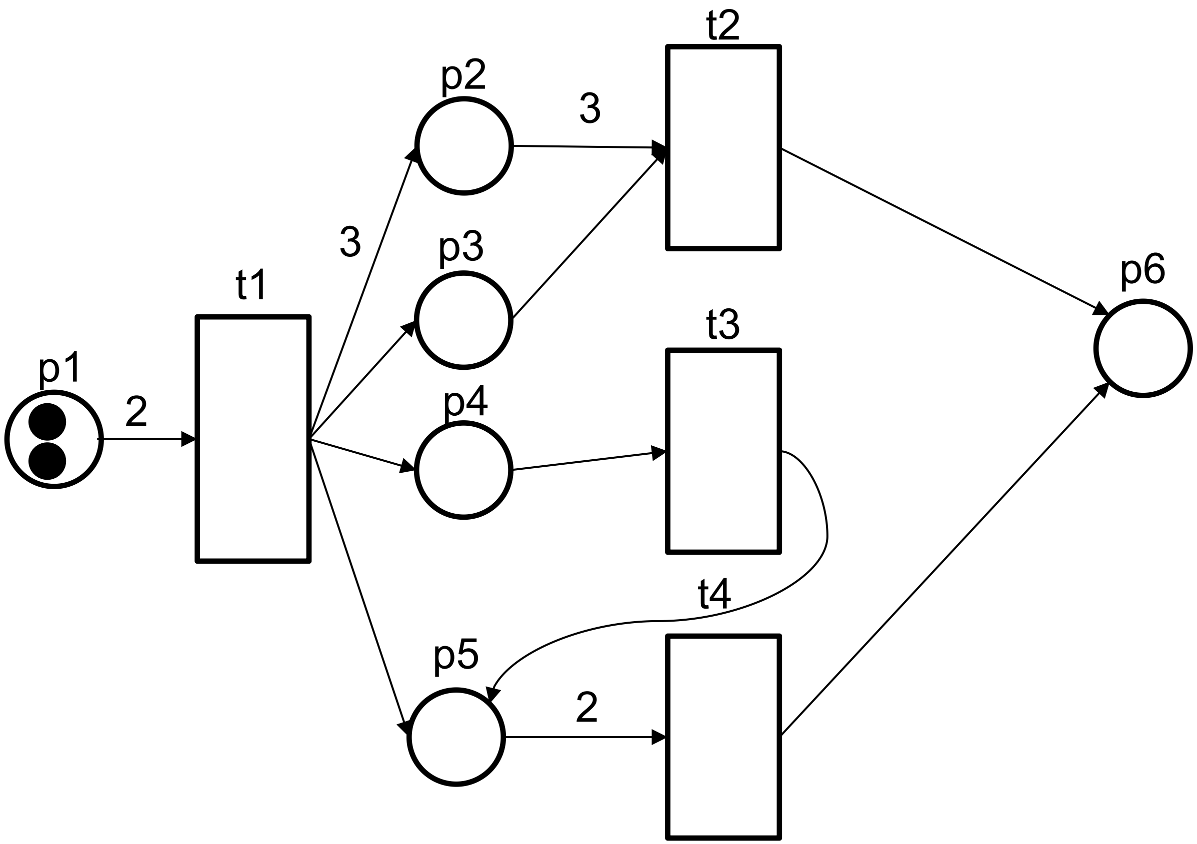
W = {{t1,p2,3}, {p1,t1,2}, {p5,t4,2}, {p2,t2,3}}

M_0 = {{p1,2}}

Petri Nets

What type of Petri Net is this and why? Draw the Petri Net next.

PT Net



Petri Nets

$N = \{P,T,F,C,W,M_0\}$
P = {p1,p2,p3,p4,p5,p6}
T = {t1,t2,t3,t4,t5}
F = {{p1,t1} , {t1,p2} , {p2,t2} ,{t1,p3}, {t1,p4}, {t1,p5}, {p3,t2} , {t2,p6}, {p4,t3}, {t3,p5}, {p5,t4} ,{t4,p6}}
C = {}
W = {{t1,p2,3},{p1,t1,2},{p5,t4,2},{p2,t2,3}}
M_0 = {{p1,2}}

What does C stand for and what is the value for p3?

What does W stand for and what is the value for {t2,p6}?

Is the net live?

Petri Nets

$N = \{P,T,F,M_0\}$

P = {p1,p2,p3,p4,p5,p6}

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C = {}

W = {{t1,p2,3},{p1,t1,2},{p5,t4,2},{p2,t2,3}}

M_0 = {{p1,2}}

What does C stand for and what is the value for p3?

C = Capacity = Infinite in this case

What does W stand for and what is the value for {t2,p6}?

W = Weight = 1 is standard

Is the net live?

No